

Machine Learning

7. *Supervised Learning* 3-2

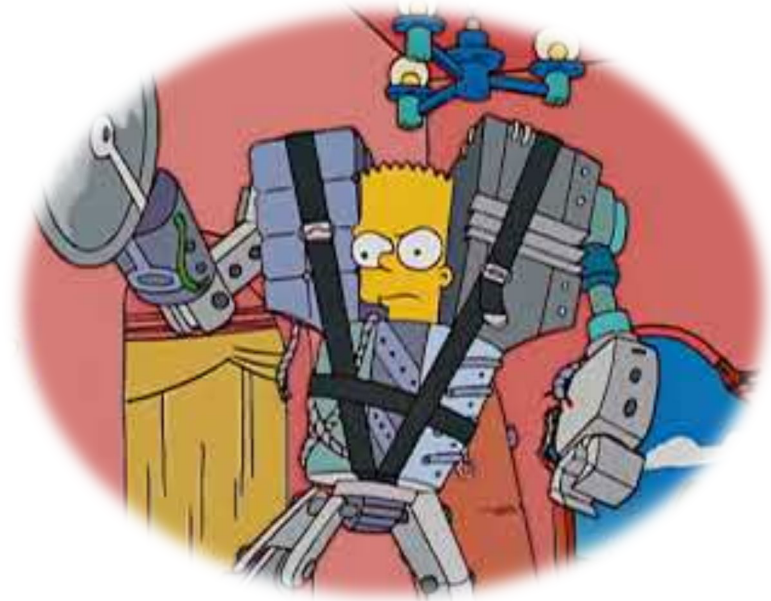
Sunwoo Lee
sunwool@inha.ac.kr



INHA UNIVERSITY

Today, we will learn...

- Supervised Learning 3-2
 - Soft-Margin SVM



Do not share or distribute these slides!

저작권 안내 / Copyright Notice

본 사이트에서 수업 자료로 이용되는 저작물은 저작권법 제25조 수업목적저작물 이용 보상금제도의 의거, 한국복제전송저작권협회와 약정을 체결하고 적법하게 이용하고 있습니다. **약정범위를 초과하는 사용은 저작권법에 저촉될 수 있으므로 수업자료의 대중 공개·공유 및 수업 목적외의 사용을 금지합니다. (일부 공개로 제공되는 유튜브 강의를 불법으로 공유하거나 배포하는 행위를 금함)**

The teaching materials provided on this site are copyrighted and signed legally by the Korea Reproduction and Transmission Rights Association. Using these materials beyond the scope of the agreement is a violation against copyright laws. **Hence, one is not allowed to open or share the materials in public. Using them outside of the class is strictly prohibited.**

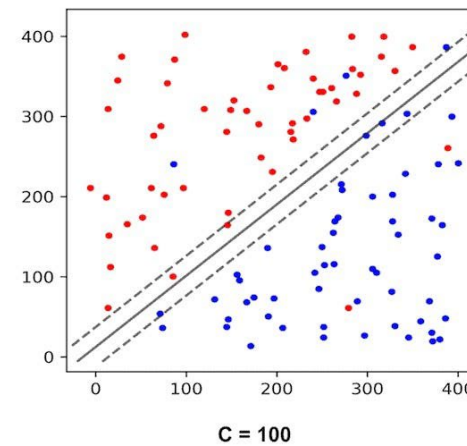
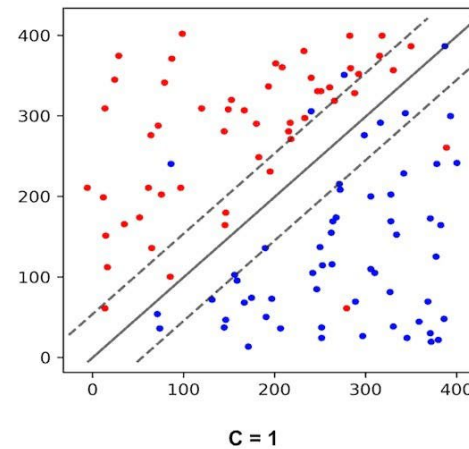
2020. 3. 11.

인하대학교·한국복제전송저작권협회

Inha University · Korea Reproduction and Transmission Rights Association

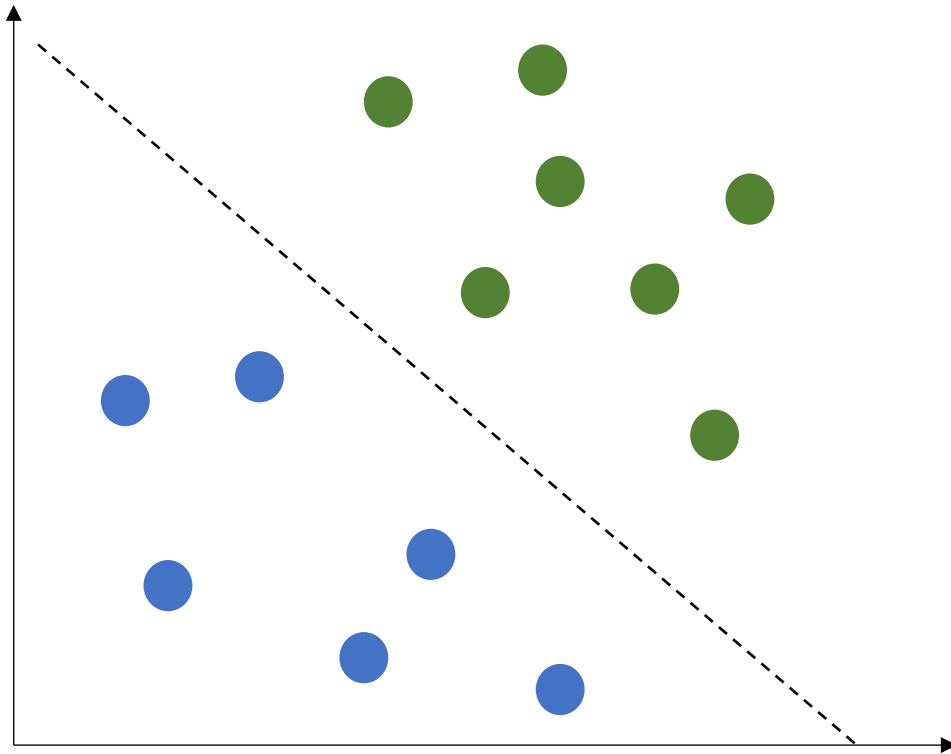
- ***Soft-margin Support Vector Machine (SVM)***

SVM Parameter C

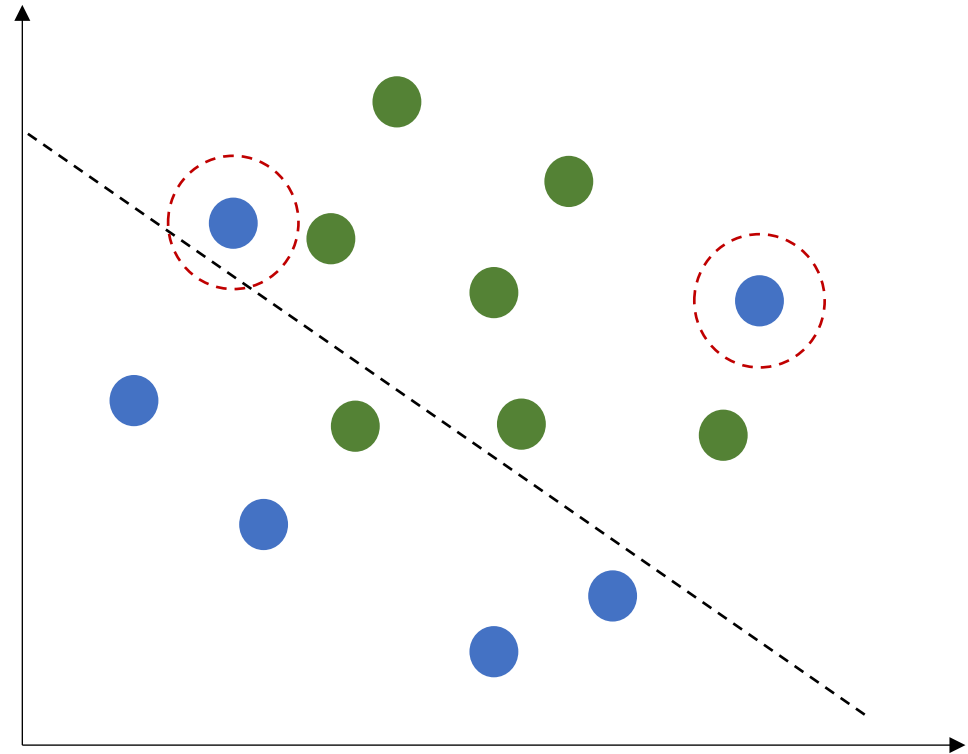


Linearly Non-Separable Problems

What if it is not possible to clearly separate the data into different groups?



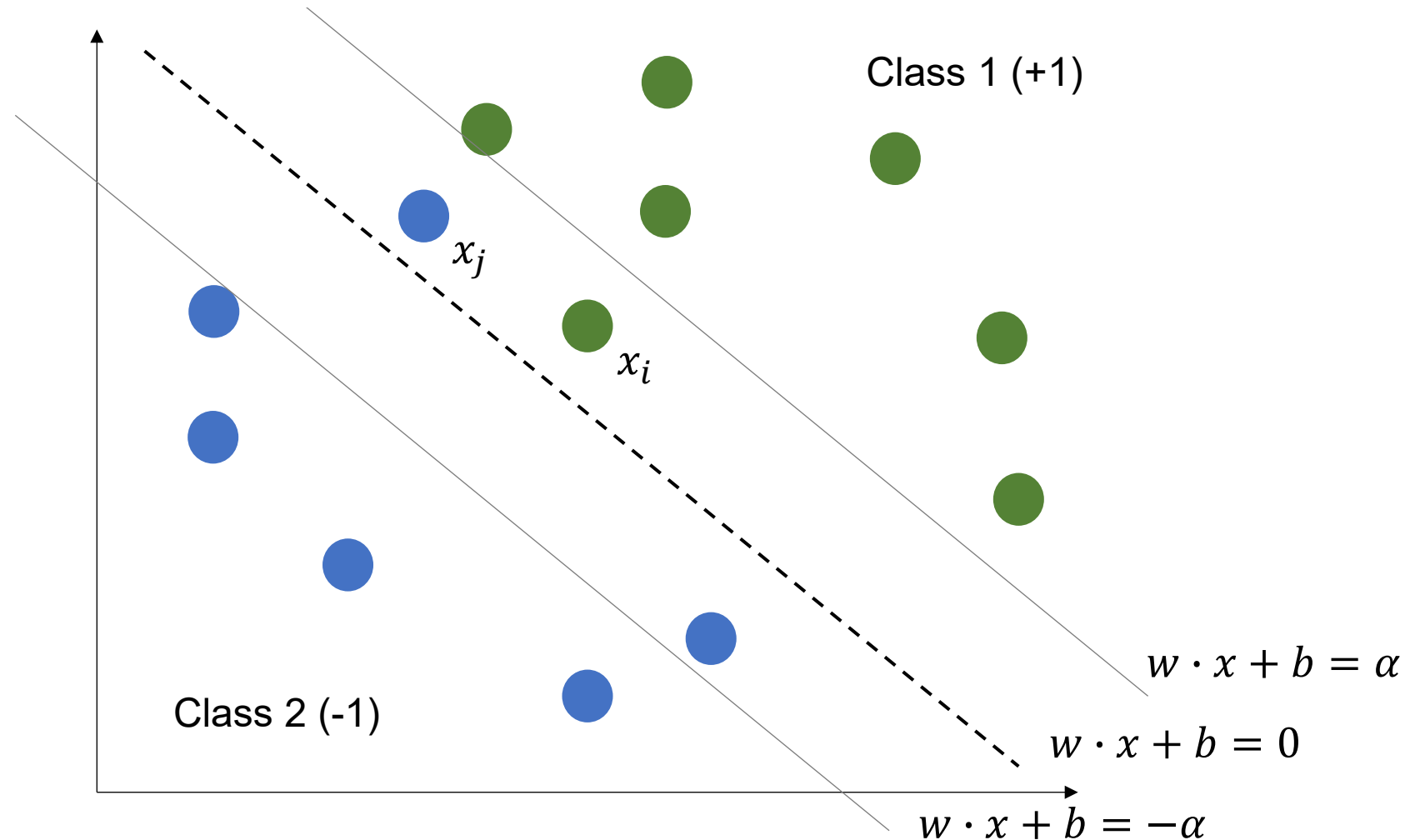
Linearly Separable



Linearly Non-Separable

Linearly Non-Separable Problems

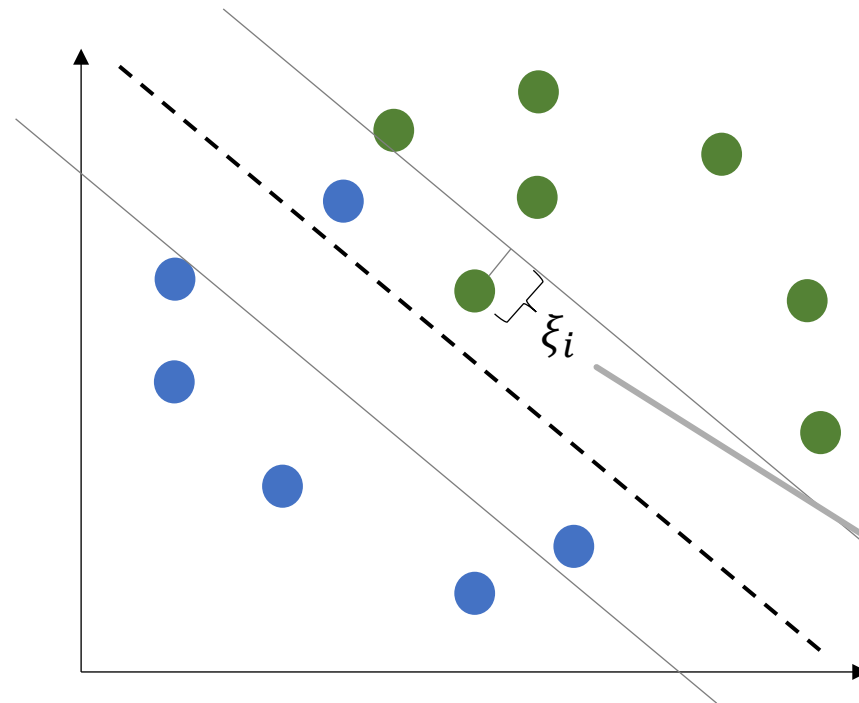
Why not just allow 'slightly' wrong classifications?



Main Idea of Soft-Margin

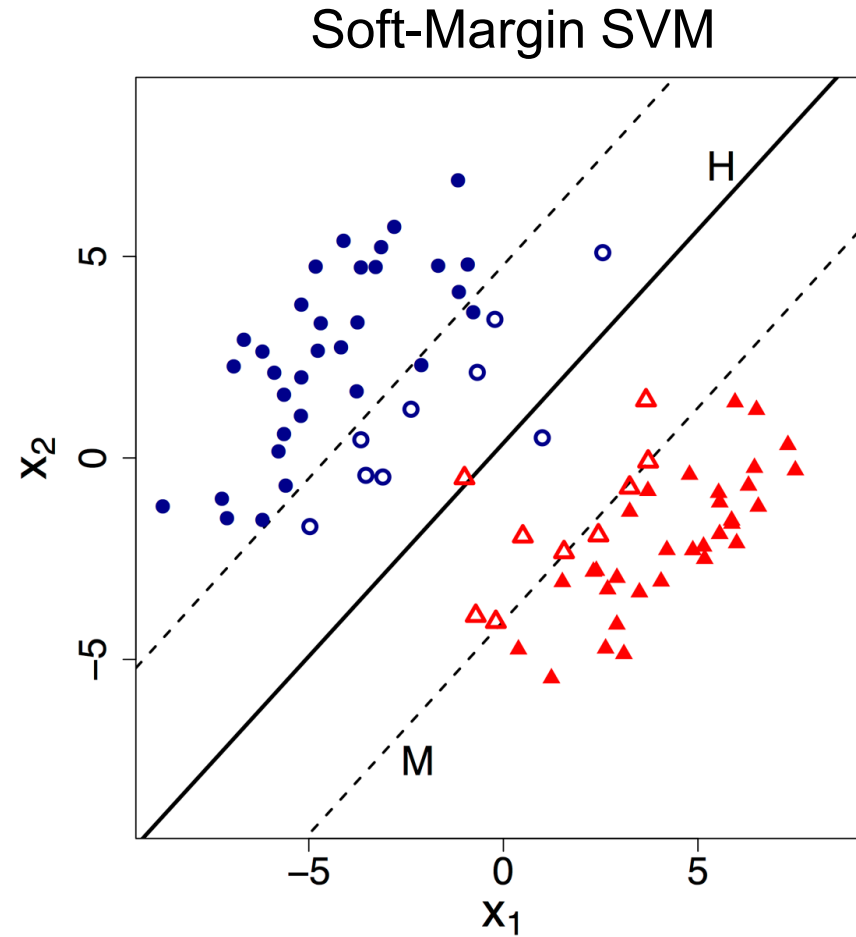
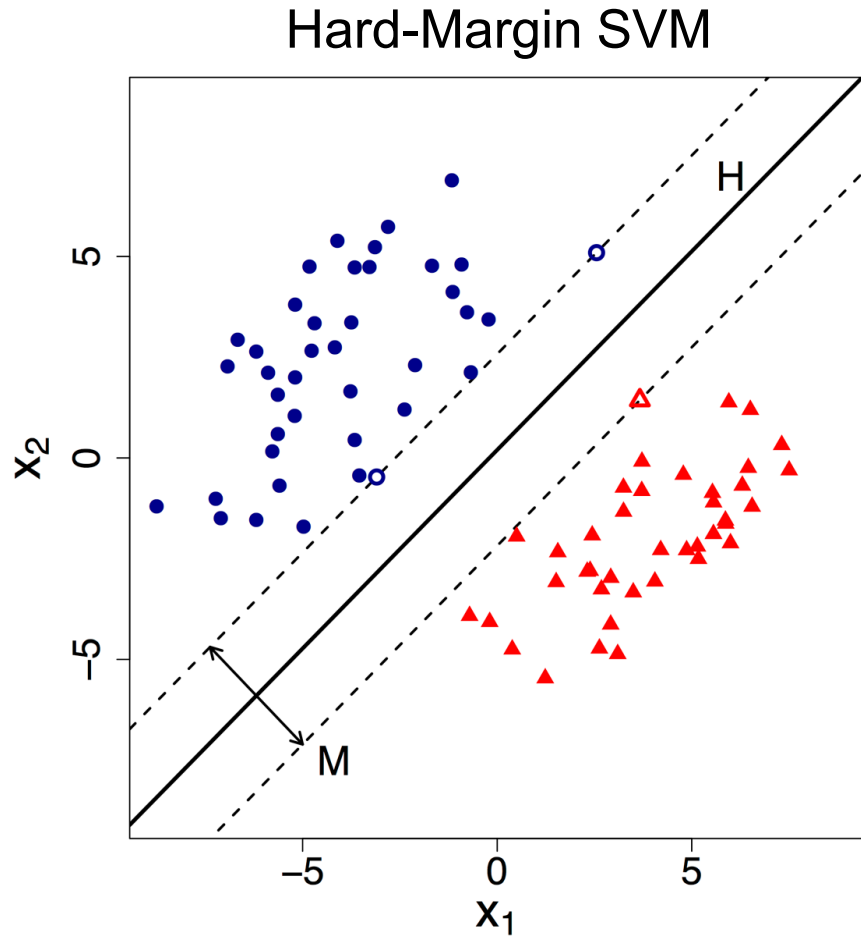
Allow a certain 'small' degree of error!

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n$$



Error
(넘어가 있는 정도)
: 이 정도의 오류는 허용
하면 문제 해결 가능

Soft-Margin SVM



일반적으로 더
많이 쓰임

Problem Formulation

Original Problem

Regularization term

$$\begin{aligned} &\text{Minimize } \frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i \\ &\text{Subject to } y_i(w^T x_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n \end{aligned}$$

- Introduce a new decision variable ξ , which refers to the soft margin. 일정 수준의 오류는 허용하되 오류는 가능한 최소화
- C is the factor of the regularization term, which determines how much error will be allowed. 라그랑지안의 경우와 달리 C 는 하이퍼파라미터, 즉 학습 중에서는 고정된 상수임
=> 규제(오류율에 패널티 부여)를 위한 것

Impact of C

Regularization term

Minimize $\frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i$
Subject to $y_i(w^T x_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n$

Original Problem

- C is a hyper-parameter that determines the training error.
 - Large C : strong constraints $\rightarrow$ will not allow a high degree of error.
 - Small C : weak constraints $\rightarrow$ will allow a high degree of error.
- By having the regularization term, we can find the separator even when the data is not linearly separable!

Lagrangian Formulation (1/3)

Original Problem

Regularization term

$$\begin{aligned} &\text{Minimize } \frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i \\ &\text{Subject to } y_i(w^T x_i + b) \geq 1 - \xi_i, \text{ and } \xi_i > 0 \quad i = 1, 2, \dots, n \end{aligned}$$



Lagrangian Primal

$$\begin{aligned} &\min_{w, b, \xi} \left(\frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n a_i (y_i(w^T x_i + b) - 1 + \xi_i) - \sum_{i=1}^n \gamma_i \xi_i \right) \\ &\text{Subject to } a_i \cdot \gamma_i \geq 0, \quad i = 1, 2, \dots, n \end{aligned}$$

Lagrange multipliers

Lagrangian Formulation (2/3)

Lagrangian Primal

$$\min_{w,b,\xi} \left(\frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n a_i (y_i (w^T x_i + b) - 1 + \xi_i) - \sum_{i=1}^n \gamma_i \xi_i \right)$$

Subject to $a_i \cdot \gamma_i \geq 0, \quad i = 1, 2, \dots, n$

- First, we solve the primal ($\min_{w,b,\xi} \dots$).

$$L = \frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n a_i (y_i (w^T x_i + b) - 1 + \xi_i) - \sum_{i=1}^n \gamma_i \xi_i$$

$$\frac{\partial L}{\partial w} = w - \sum \alpha_i y_i x_i = 0 \quad \Rightarrow \quad w = \sum \alpha_i y_i x_i$$

$$\frac{\partial L}{\partial b} = -\sum \alpha_i y_i = 0 \quad \Rightarrow \quad \sum \alpha_i y_i = 0$$

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \gamma_i \quad \Rightarrow \quad C - \alpha_i - \gamma_i = 0, \forall i$$

Lagrangian Formulation (3/3)

- Then, we solve the dual by plugging what we found from primal to the Lagrangian.

$$\begin{aligned}
 L &= \frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1 + \xi_i) - \sum_{i=1}^n \gamma_i \xi_i \\
 &= \frac{\|w\|^2}{2} - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1) + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i \xi_i - \sum_{i=1}^n \gamma_i \xi_i \\
 &= \frac{\|w\|^2}{2} - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1) + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n (\alpha_i + \gamma_i) \xi_i \\
 &= \frac{\|w\|^2}{2} - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1) + C \sum_{i=1}^n \xi_i - C \sum_{i=1}^n \xi_i \\
 &= \frac{\|w\|^2}{2} - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1) \\
 &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j
 \end{aligned}$$

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \gamma_i \Rightarrow C - \alpha_i - \gamma_i = 0, \forall i$$

Lagrangian Dual

- The dual ($\max_{\alpha}$...) becomes as follows.

$$L = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j$$
$$s. t. \underbrace{\sum_{i=1}^n \alpha_i y_i = 0}_{\frac{\partial L}{\partial b} = 0} \text{ and } \underbrace{C - \alpha_i - \gamma_i = 0}_{\frac{\partial L}{\partial \xi_i} = 0}, \quad i = 1, 2, \dots, n$$

Note that $\alpha_i \geq 0$ and $\gamma_i \geq 0$, and thus $0 \leq \alpha_i \leq C$.

Lagrangian Dual Comparison

Soft-Margin SVM

$$L = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j$$
$$\text{s.t. } 0 \leq \alpha_i \leq C, \quad i = 1, 2, \dots, n$$

Only difference here!

Hard-Margin SVM

$$L = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j$$
$$\text{s.t. } 0 \leq \alpha_i, \quad i = 1, 2, \dots, n$$

Soft-Margin SVM has the same Lagrangian dual as Hard-Margin SVM!
We can solve QP problem like before...

Finally, our Soft-Margin SVM model is...

- Once we find α_i by solving the QP problem, based on the decision rule, we can find w and b .

$$w = \sum \alpha_i y_i x_i \text{ and } y_i(w^T x_i + b) = 1$$

- Then, given a value of C (hyper-parameter), we can find γ_i .

$$C - \alpha_i - \gamma_i = 0, \forall i$$

- Finally, we can find ξ_i by plugging in everything into our Lagrangian dual.

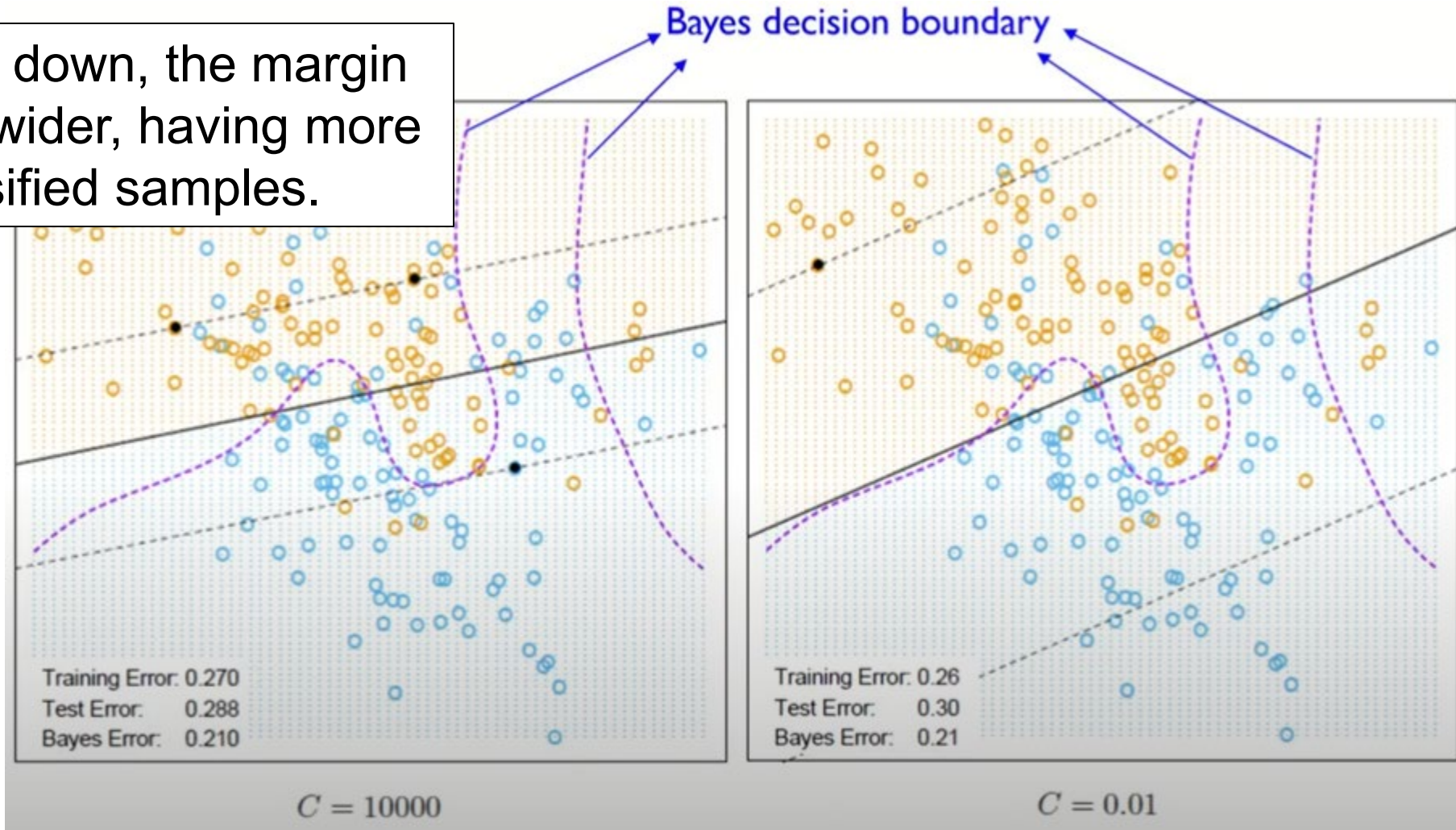
$$\frac{\|w\|^2}{2} + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i (y_i(w^T x_i + b) - 1 + \xi_i) - \sum_{i=1}^n \gamma_i \xi_i$$

Characteristics of the Solution

- From the KKT condition (complementary slackness), we can find...
 $\alpha_i(y_i(w^T x_i + b) - 1 + \xi_i) = 0$ s.t. $\alpha_i = C - \gamma_i$ and $\gamma_i \xi_i = 0, \forall i$.
 - $\alpha_i = 0 \Rightarrow \gamma_i = C \Rightarrow \xi_i = 0$
 - So, $y_i(w^T x_i + b) - 1 \neq 0$, which means x_i is an internal point.
 - $0 < \alpha_i < C \Rightarrow \gamma_i > 0 \Rightarrow \xi_i = 0$
 - So, $y_i(w^T x_i + b) - 1 = 0$, and x_i is a boundary point (support vector).
 - $\alpha_i = C \Rightarrow \gamma_i = 0 \Rightarrow \xi_i > 0$
 - So, $\alpha_i(y_i(w^T x_i + b) - 1 + \xi_i) = -\alpha_i \xi_i \neq 0$, and x_i is beyond the boundary.

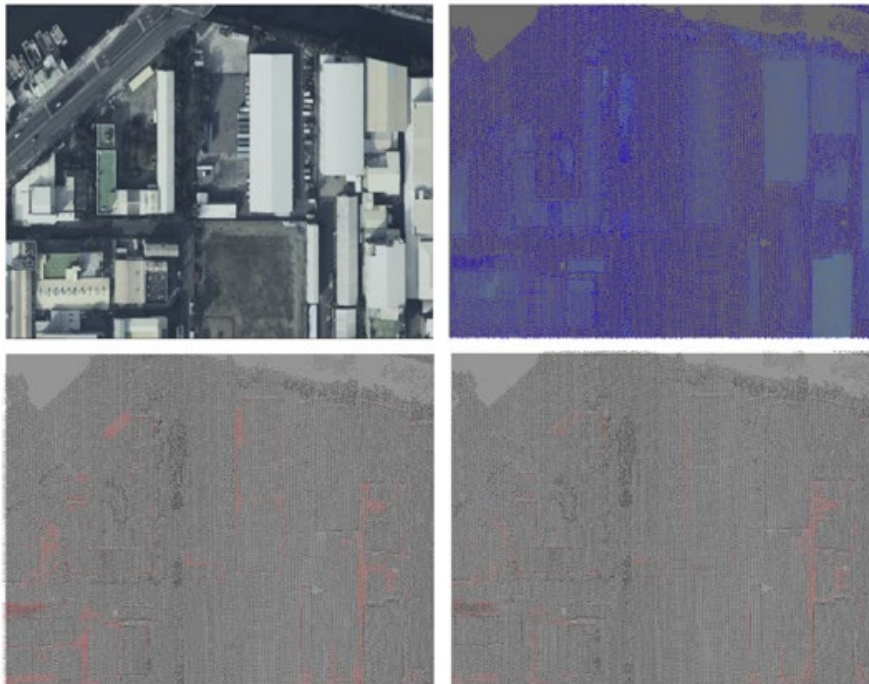
Example

As C goes down, the margin becomes wider, having more miss-classified samples.



Applications of Soft-Margin SVM (1/3)

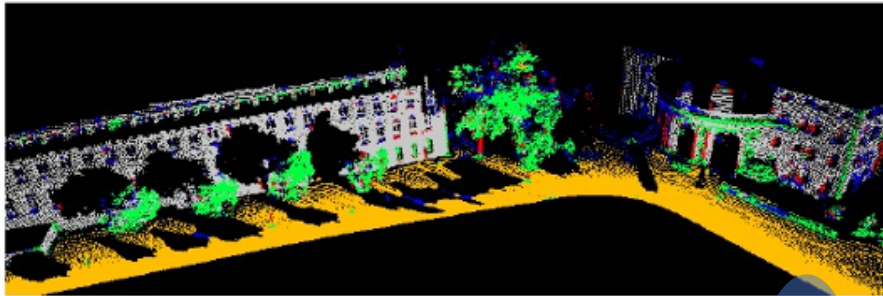
- [1] Eigen-feature analysis of weighted covariance matrices for LiDAR point cloud classification (Lin et al., 2014)



Top left: orthoimage; top right: point clouds;
Bot left: classification results from standard eigen features
Bot right: classification results from generated features

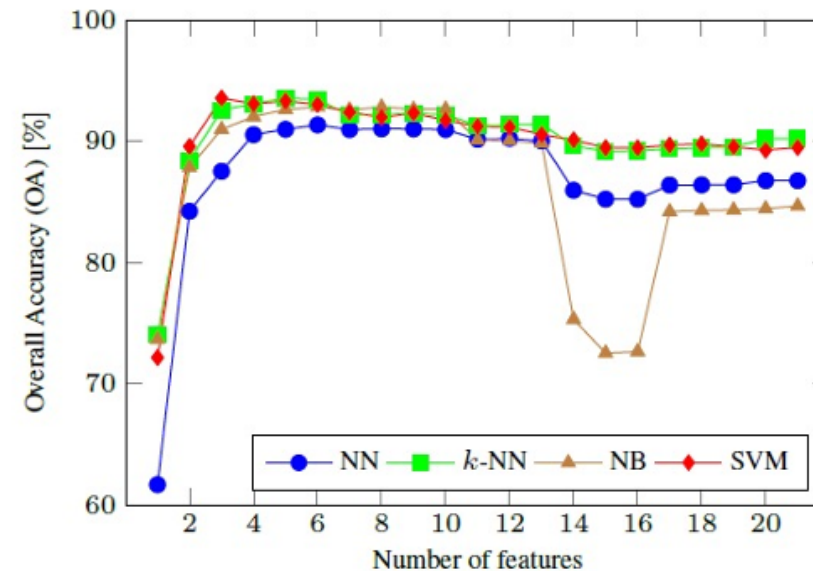
Applications of Soft-Margin SVM (2/3)

- [2] Feature relevance assessment for the semantic interpretation of 3D point cloud data (Weinmann et al., 2013)



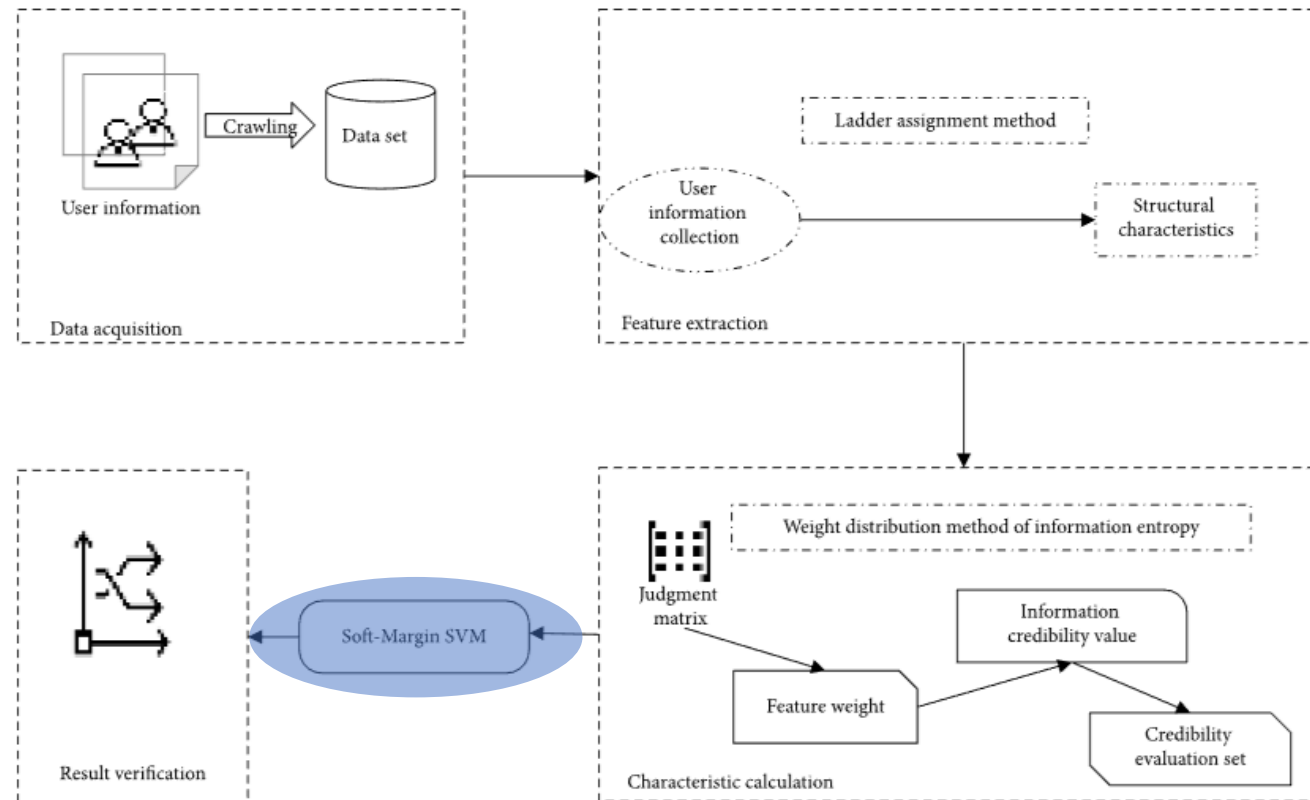
	NN	k -NN	NB	SVM
all features	86.74	90.21	84.66	89.48
only 3D features	85.70	88.64	85.80	88.34
only 2D features	82.66	83.80	77.09	80.06
eigenvalue-based 3D features	80.36	82.42	83.92	84.81
$\{L_\lambda, P_\lambda, S_\lambda\}$	62.91	75.72	76.90	75.26
five best-ranked features	90.97	93.53	92.61	93.32

Table 3: Overall classification accuracies in % for different feature subsets and different classifiers.



Applications of Soft-Margin SVM (3/3)

- [3] S3UCA: Soft-Margin SVM-based Social Network User Credibility Assessment Method (Zhao et al., 2021)



Soft-Margin SVM vs. Kernel Trick

- Both approaches have an extra computational cost.
 - Soft-Margin SVM: the extra regularization term ξ_i .
 - Kernel trick: the kernel function $K(x_i, x_j)$.

사용이 매우 간단하고 일반화 성능이 뛰어나나 정확도 자체는 다소 희생됨

- Soft-Margin SVM finds a linear solution that not only maximizes the margin but also minimizes the error of the outliers.
- Kernel finds a non-linear solution that perfectly classifies the data. 훈련 정확도가 뛰어나나 계산 비용이 크고 모델 복잡도가 증가하고 과적합 위험성이 큼
- It's a matter of choice again!



Any Questions?

That's it! We covered all SVMs!

